

STAT 1210: Assignment 1

Part I: Exploring Data

Fall Semester 2026

Instructions

- Your work should show all steps clearly for each question.
- Your solutions must be handwritten on lined or letter sized paper. Your name and student number must appear on the first page.
- Your name should also appear on every additional page and all pages should be stapled together.
- R is the preferred programming language, although equivalent Python code is also acceptable unless a question specifically requires R output.
- Programming resources, textbooks and software manuals may be consulted for coding tasks, but all submitted work must be your own. Solutions generated by generative AI tools are not permitted.

Question 1: Choosing and Evaluating a Categorical Display

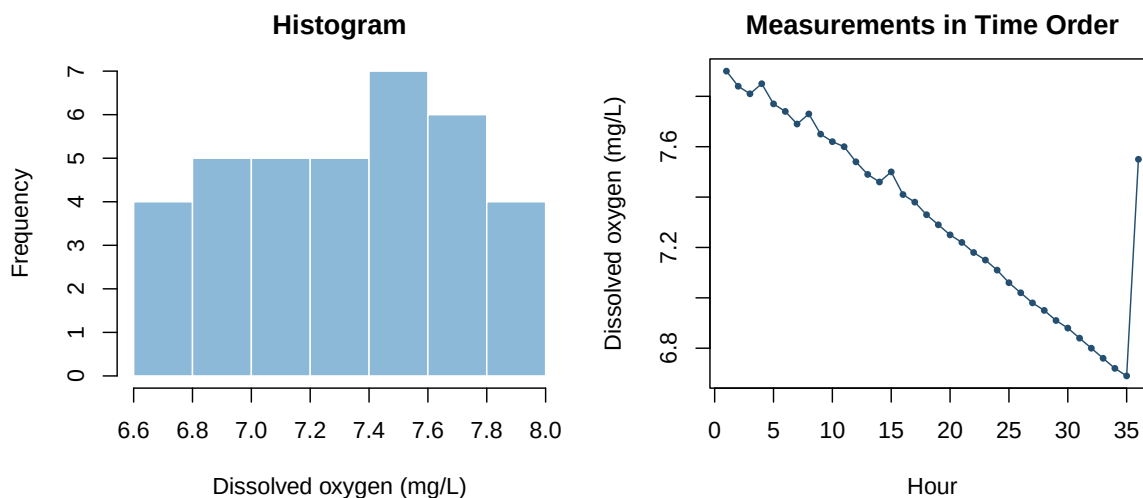
An environmental laboratory compares the germination of seeds collected from three habitats. The numbers of seeds that germinated within 14 days are shown below.

Habitat	Seeds tested	Seeds germinated
Coastal dunes	40	30
Forest edge	75	48
Restored field	120	84

1. Identify the **individuals** and the two principal variables recorded for each seed. Classify each variable as categorical or quantitative. If a variable is quantitative, state whether it is discrete or continuous.
2. Calculate the germination percentage for each habitat. Which habitat has the largest number of germinated seeds, and which has the highest germination rate? Explain why these answers are not the same.
3. A researcher proposes a pie chart with slices representing the counts of germinated seeds (30, 48, and 84) from each habitat. Explain why this display would not provide a fair comparison of germination success among the habitats.
4. Suggest an appropriate type of graph for comparing the three germination rates. Specify what should appear on each axis and explain one design choice that would prevent the graph from exaggerating the differences between the habitats.

Question 2: A Distribution Viewed in Two Ways

A sensor recorded dissolved oxygen in a freshwater tank every hour for 36 hours. The two displays below contain exactly the same observations.



1. Describe the histogram using **SOCS**: shape, possible outliers, approximate center and spread. Clearly indicate which of these features are difficult to judge from the histogram alone.
2. What important pattern is visible in the time plot but largely hidden by the histogram? Explain why the time plot is the more useful first display for assessing whether the tank conditions remained stable.
3. The final observation was recorded immediately after the sensor was recalibrated. Is it automatically appropriate to delete this observation? Describe two reasonable checks that a researcher should make before deciding how to handle it.
4. Suppose the researcher recreated the histogram using only four very wide bins. Explain how this choice might change the apparent shape of the distribution without changing any of the underlying observations.

Question 3: Resistance of Measures of Center and Spread

The following values are the time, in minutes, required for 11 samples to complete a chemical reaction:

12.1, 12.4, 12.7, 12.8, 13.0, 13.1, 13.3, 13.5, 13.8, 14.0, 31.6.

1. Use the following one-line R commands to calculate the mean, median, standard deviation and IQR:

```
reaction_time <- c(12.1, 12.4, 12.7, 12.8, 13.0, 13.1,
                  13.3, 13.5, 13.8, 14.0, 31.6)
mean(reaction_time)
median(reaction_time)
sd(reaction_time)
IQR(reaction_time)
```

2. A review of the laboratory record confirms that 31.6 is a transcription error and should have been 13.6. Replace this value in your R code and recalculate the four statistics using the same commands.
3. Compare the changes in the mean and median. Then compare the changes in the standard deviation and IQR. Use the concept of resistance to explain the pattern you observe.
4. Before the error was discovered, which pair of statistics would have provided a more representative summary of the reaction-time distribution: the mean and standard deviation, or the median and IQR? Justify your choice using the dataset and your calculated results.

Question 4: Boxplots, Outlier Fences and Scientific Interpretation

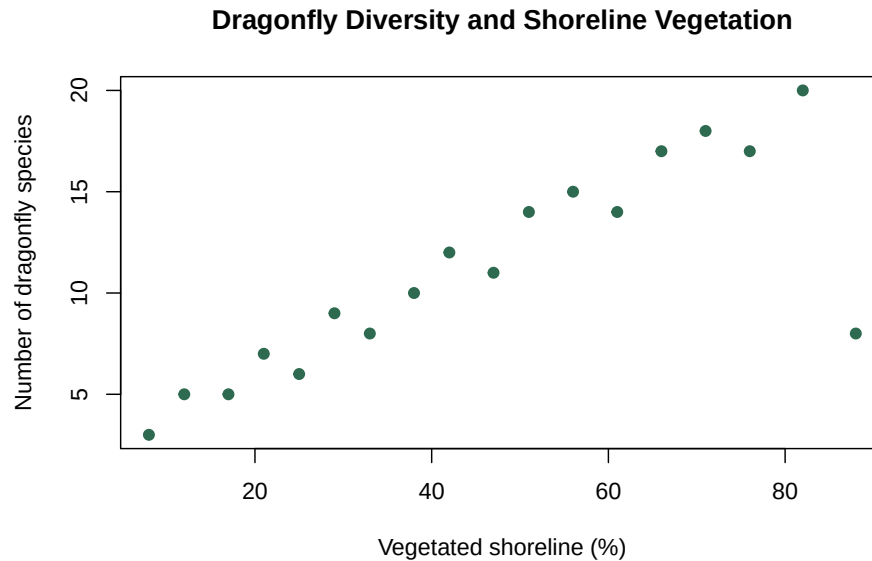
Two methods were used to measure the concentration of the same compound. Their summaries, in micromoles per litre, are:

Method	Minimum	Q_1	Median	Q_3	Maximum	Mean	Standard deviation
A	18.2	19.4	20.1	20.8	22.0	20.1	0.9
B	12.5	18.9	20.0	21.2	28.4	20.2	3.6

1. Calculate the IQR and the lower and upper ($1.5 \times \text{IQR}$) fences for each method.
2. Using only the supplied summaries, determine whether the minimum or maximum of either method must be flagged as a suspected outlier. Show your comparisons with the fences.
3. Compare the two methods in terms of center and variability. Which method appears more consistent? Support your answer with at least two numerical comparisons.
4. Explain why the similar means do not imply that the methods perform similarly. Which measures of center and spread would you emphasize for Method B, and why?
5. A technician recommends deleting both extreme Method B measurements solely because they fall beyond the outlier fences. Explain why the ($1.5 \times \text{IQR}$) rule alone is not enough to justify deleting observations.

Question 5: Interpreting a Scatterplot

An ecologist studied 18 ponds. For each pond, the ecologist recorded the percentage of shoreline covered by vegetation and the number of dragonfly species observed during a standardized survey.



1. Identify the explanatory and response variables and explain why these roles are reasonable for this ecological question.
2. Describe the overall association in terms of direction, form, and strength.
3. Identify the observation that departs most clearly from the overall pattern. Explain why it is a bivariate unusual observation even if each of its two values might not be unusual when considered separately.
4. Predict qualitatively how removing this observation would affect the correlation: would its magnitude probably become larger, become smaller, or remain about the same? Explain your reasoning based on the scatterplot.
5. Can this observational study establish that increasing shoreline vegetation causes greater dragonfly diversity? Explain your answer and identify one plausible lurking variable.

Question 6: Correlation Can Miss a Strong Pattern

A scientist measures a response at seven equally spaced values of an explanatory variable:

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

1. Draw a scatterplot by hand or generate one using the single R command below:

```
plot(c(-3,-2,-1,0,1,2,3), c(9,4,1,0,1,4,9))
```

Describe the direction, form, and strength of the relationship.

2. Use the following R command to calculate Pearson's correlation:

```
cor(c(-3,-2,-1,0,1,2,3), c(9,4,1,0,1,4,9))
```

Interpret the value of (r) specifically as a measure of **linear** association.

3. A classmate concludes, "Because r is exactly zero, there is no relationship between x and y ". Explain precisely why this conclusion is incorrect.
4. Suppose x is converted to a different unit of measurement using the formula $x_{\text{new}} = 10x + 50$. Without recalculating, state what happens to r and explain why.
5. What general rule about correlation and scatterplots does this example illustrate?

Question 7: Least-Squares Regression

A horticulture student investigates whether daily light exposure can be used to predict the dry biomass of seedlings after four weeks. Five experimental groups produced the following averages:

Daily light, x (hours)	1	2	3	4	5
Dry biomass, y (g)	2	4	5	4	5

For these data, the following sample statistics have been calculated:

$$\bar{x} = 3, \quad \bar{y} = 4, \quad s_x = 1.581, \quad s_y = 1.225, \quad r = 0.775.$$

Use

$$b = r \frac{s_y}{s_x}, \quad a = \bar{y} - b\bar{x}, \quad \text{and} \quad \hat{y} = a + bx$$

for the hand calculations in Parts 1–3.

Part 1: Construct and Interpret the Regression Line Manually

1. Calculate the slope b and intercept a .
2. Write the least-squares regression equation.
3. Interpret the slope in context, including its units.
4. Interpret the intercept in context. Comment on whether this intercept is scientifically useful given the observed range of light exposure.

Part 2: Prediction and Residual by Hand

1. Use your equation to predict dry biomass for a group receiving 4 hours of light per day.
2. The observed biomass at 4 hours is 4 g. Calculate the residual using

$$e = y - \hat{y}.$$

3. Interpret the sign and magnitude of this residual in context.

Part 3: Strength and Limitations by Hand

1. Calculate the coefficient of determination (R^2).
2. Interpret R^2 in context. Your interpretation must refer to the observed variation in dry biomass and the fitted linear relationship with daily light exposure.
3. The student wants to use the fitted line to predict biomass for 8 hours of light per day. Is this interpolation or extrapolation? Explain why this prediction should be treated cautiously.

Part 4: Reproduce the Analysis in R

Enter the data and fit the model using:

```
light <- c(1, 2, 3, 4, 5)
biomass <- c(2, 4, 5, 4, 5)

model <- lm(biomass ~ light)
model
summary(model)
predict(model, newdata = data.frame(light = 4))
residuals(model)
```

Using the R output:

1. Report the estimated intercept and slope and compare them with your hand calculations from Part 1.
2. Report R's predicted biomass at 4 hours and the corresponding residual. Compare them with Part 2.
3. Report the multiple R^2 value and compare it with Part 3.
4. In two or three sentences, explain what this comparison demonstrates about using `lm()` instead of performing every regression calculation by hand.

Bonus Question: How R Calculates Quartiles

Different statistical software packages use different algorithms to calculate quartiles. In fact, the R `quantile()` function has 9 different methods built in! By default, R uses an **interpolation method** (Type 7) rather than simply picking a single middle value from the dataset.

To find a specific percentile p (where $p = 0.25$ for Q_1 , and $p = 0.75$ for Q_3) for a dataset of size n , R uses the following steps:

1. **Find the exact index position (h):**

$$h = (n - 1)p + 1$$

2. **Split the index:** Separate h into its integer part (i) and its fractional part (f).

3. **Interpolate:** Using the sorted values x , calculate the quantile:

$$Q = x_i + f \times (x_{i+1} - x_i)$$

Consider the following small dataset of $n = 6$ sorted observations:

10, 14, 18, 24, 30, 42

You may run the following R code to find the quartiles.

```
## 25% 75%  
## 15.0 28.5
```

1. Use the provided formulas to calculate the exact index position (h) for Q_1 ($p = 0.25$). Then, use interpolation to calculate Q_1 .
2. Repeat the process to calculate Q_3 ($p = 0.75$).
3. Do your manual calculations match R's default output? Explain briefly why an interpolation method might provide a better summary of a small dataset than simply choosing the nearest observed data point.

End of assignment